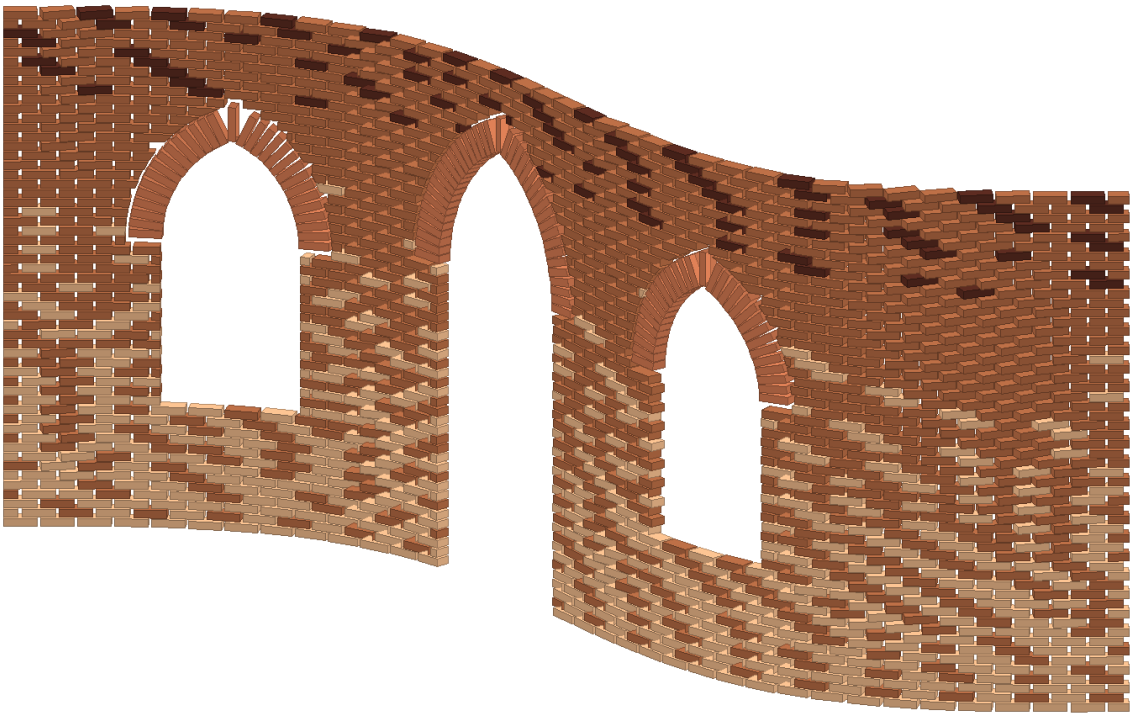


Hezarbaaf

هزارباف

Parametric brickwork for Grasshopper · component guide and tutorials

Version 0.9 · Rhino 8 · English guide with Persian component names
Every picture in this guide was produced by Hezarbaaf's own geometry engine, the same code the plugin runs.

Contents

Getting started
install, your first wall in one minute, key ideas

Showcase
a curved parametric wall with arched openings, step by step

Part 1 · The Hezarbaaf tab: the essentials
Presets, Brick Wall, Field Attractor, Field Wave, Brick Operation, Brick Output, Mortar, Bake Blocks

Part 2 · Walls
Quick Façade, Corner Wall, Loft Wall, Section Wall, Surface Wall, Tower, Arched Openings, Details

Part 3 · More fields
Field Ripple, Field Noise, Field Bitmap, Field Combine, Field Preview

Part 4 · Colour and material
Brick Tones, Glass Bricks

Part 5 · Hezarbaaf motifs
Hezarbaaf Wall, Banna'i Motif, Image Motif

Part 6 · From design to site
Raj Sheet, Quantities, Rotation Stencils, Assembly Order, Elevation Drawing

Part 7 · Ladybug bridge
Ladybug Geometry, Ladybug Results

Part 8 · Library
Brick Formats, Bonds, Language

Troubleshooting
the common mistakes and what the messages mean

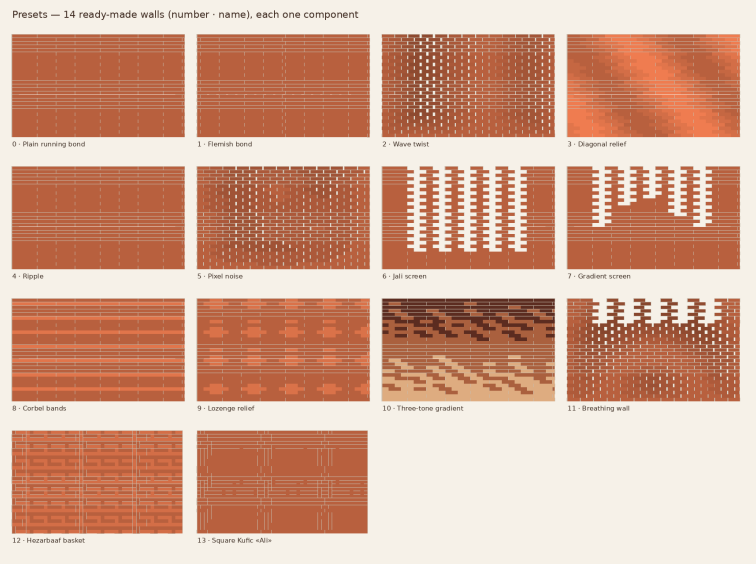
Getting started

Install

1. Copy **Hezarbaaf.gha** and **Masonry.Core.dll** into Grasshopper's Components folder (Grasshopper: *File › Special Folders › Components Folder*).
2. On Windows, right-click each file › Properties › tick **Unblock**.
3. Restart Rhino 8. Two new tabs appear: **Hezarbaaf** (the essentials) and **Hezarbaaf Advanced** (everything else).

Your first wall in one minute

1. Work in a Rhino file in **metres**. Draw a line about 4 m long.
2. In Grasshopper, add a **Curve** parameter, right-click › *Set one Curve*, and pick the line.
3. From the Hezarbaaf tab add **Presets** and **Brick Output**. Wire Curve → Presets · C, and Presets · W → Brick Output · W.
4. Change Presets · P from 0 to 13 to see 14 different walls.

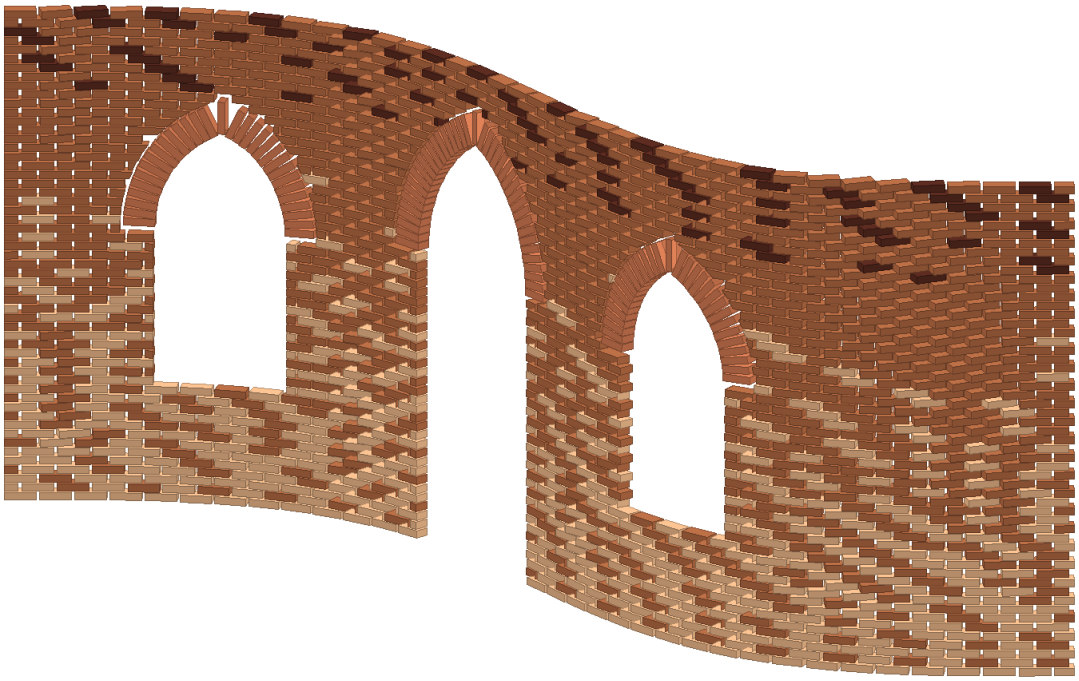


The 14 presets on a 3 m wall.

Key ideas

- **Wall data (W).** Most components pass a *wall* to each other through the W input and output. Start with Brick Wall, Presets or another wall component; operations change the wall; Brick Output, Bake and the site components read it.
- **Fields (Fd).** A field is a number from 0 to 1 over the wall's face. It tells an operation where to act and how strongly. Combine fields to make complex designs from simple ones.
- **Steps.** Operations snap to a few steps (e.g. 4 angles), so a mason only needs a few templates and the wall stays buildable.
- **Colours.** Red = buildability problem · light = cut brick · dark = turned · orange = projecting · purple = tapered · deep brown = arch · grey-brown = sill/coping · light blue = glass.
- **Units.** Inputs are in your Rhino model units (metres in this guide). Brick sizes and joints are always in millimetres.
- **Language.** Use the Language component (fa / en) and restart Rhino for Persian names. Persian names also work as input values, e.g. a bond name.

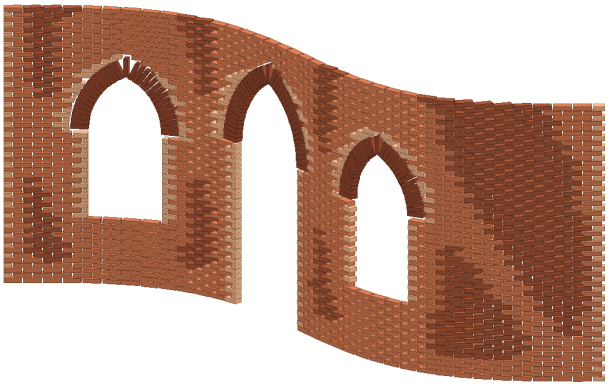
Showcase: a curved parametric wall with arched openings



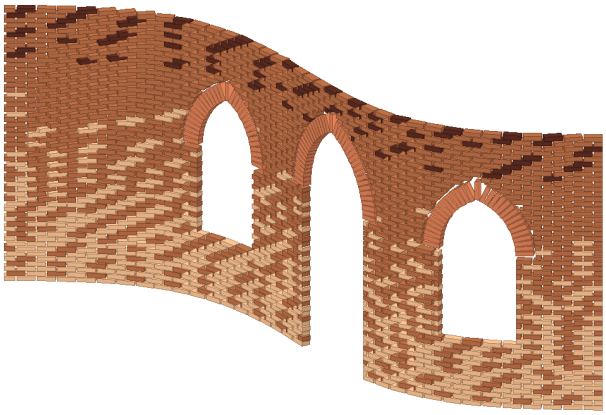
8 m S-shaped wall, 3.4 m high: three Persian pointed openings (two windows and a door), bricks turning up to 30° in diagonal waves that calm down around the openings and the ends, and three brick colours from light at the base to dark at the top. 1676 bricks, **0 buildability problems**, 5 rotation stencils.

Build it step by step

1. **The curve.** Draw an S-shaped curve about 8 m long in plan (in Grasshopper: a *Graph Mapper* or a sine through points, amplitude 0.7 m). Set it in a Curve parameter.
2. **The wall.** Brick Wall: C = curve, H = 3.4, F = *210x100x50 j25/12*. The wider 25 mm head joint leaves room for turned bricks. No OU/OV here.
3. **The openings.** Arched Openings: C = same curve, W = the wall. OU = three intervals *1.0 To 1.9*, *3.0 To 4.2*, *5.3 To 6.2*; OV = *0.8 To 1.9*, *0 To 2.0*, *0.8 To 1.9* (a Panel with one interval per line). K = 2 (pointed), S = 0.3.
4. **The wave.** Field Wave: A = 35 (diagonal), L = 1.8.
5. **Calm around the openings.** Field Attractor: W = curve, P = one point in the middle of each opening, R = 1.2. Field Combine with I = True (invert) gives 1 away from the openings and 0 at them.
6. **Mix.** Field Combine: A = the wave, B = the inverted attractor, M = 1 (multiply).
7. **Turn the bricks.** Brick Operation: W = the arched wall, Fd = the mix, M = 0 (rotate), A = 30, S = 5.
8. **Colours.** Brick Tones: W = the turned wall, F = a Field Wave with A = 90 and a long wavelength (or Field Bitmap with a vertical gradient), N = 3.
9. **Check and build.** Brick Output (C = 1 for the colours, 0 for the checks). Then Rotation Stencils, Raj Sheet and Elevation Drawing for the site.



Check colours (Brick Output C = 0): no red. Arches deep brown, turned bricks dark.



The back of the wall: the same bricks, seen from behind.

Why it works

- The **wide head joint** (25 mm) lets bricks turn 30° without touching. Brick Output would show red if they collided.
- The **calm zone** keeps the bricks straight where the arches spring and at the wall ends, which is where masons need a clean, strong edge.
- Only **5 steps** of rotation, so the whole wall needs a handful of stencils.
- Arches keep their own colour; Brick Tones only mixes the wall bricks.

Part 1 · The Hezarbaaf tab: the essentials

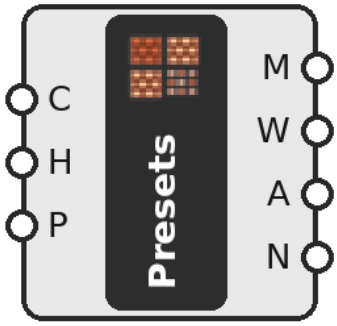
8 components: Presets, Brick Wall, Field Attractor, Field Wave, Brick Operation, Brick Output, Mortar, Bake Blocks

Presets

نمونه‌ها

Hezarbaaf › 1 Start

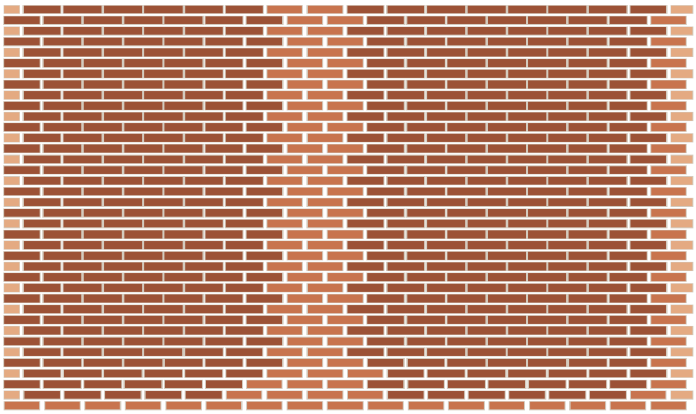
The fastest way to start. Pick one of 14 ready-made walls by number or name, give it a curve and a height, and you get a finished parametric wall. Each preset also shows one technique, so you can later rebuild it yourself with the other components.



Input	What it is	Default
C Curve	Base curve	—
H Height	Wall height	2.4
P Preset	Number (0, 1, 2 …) or name — see the Names output	2
Output	**What it is**	
M Mesh	Bricks	
W Wall	Wall data → any other Hezarbaaf component	
A About	What this preset does	
N Names	All presets (connect to a Value List or Panel)	

Try it

1. Draw a line about 4 m long in Rhino and set it in a Curve parameter.
2. Connect it to **C**. Leave **H** at 2.4.
3. Type a number in **P** (0–13), or connect a slider. Connect a Panel to **N** to see all names.
4. Connect **W** to Brick Output to see the coloured preview and the checks.



Preset 2, "Wave twist": bricks turn up to 30° following a wave, calm near the ends.

Tips

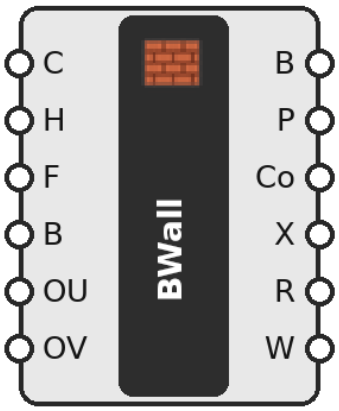
- P also accepts names in English or Persian, e.g. *Jali screen*.
- Presets 12 and 13 (Hezarbaaf basket, Kufic «Ali») use the Iranian 20×10×5 brick.

Brick Wall

دیوار آجری

Hezarbaaf › 1 Start

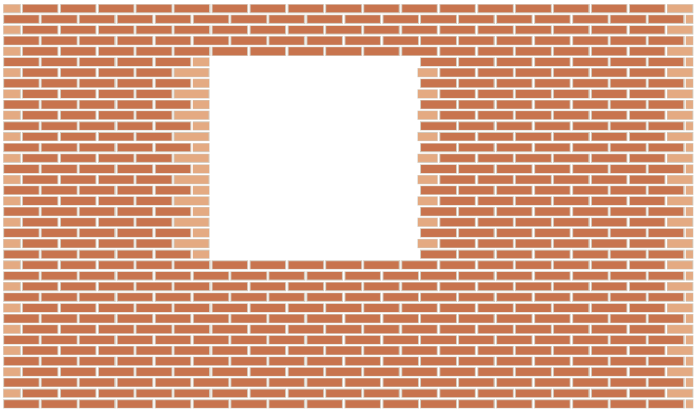
The basic wall. Lays real bricks in a real bond along any curve in plan, course by course, and cuts rectangular openings (windows, doors). Everything else in Hezarbaaf starts from a wall like this.



Input	What it is	Default
C Curve	Base curve (plan)	—
H Height	Wall height (model units)	2.0
F Format	Brick format	Waal (NL)
B Bond	Bond	Stretcher
OU Open U	Openings along the wall: start To end, e.g. a Panel with 1.2 To 2.4, or Construct Domain (not a single slider)	—
OV Open V	Opening heights: sill To top, e.g. 0.9 To 2.0 (0 To … = door)	—
Output	**What it is**	
B Bricks	Brick boxes	
P Planes	Brick planes	
Co Course	Course index	
X Cut	Cut brick	
R Report	Quantities	
W Wall	Wall data → Brick Operation / Brick Output	

Try it

1. Connect a plan curve to **C** (straight, curved or polyline) and set the height **H**.
2. Optional: type a brick in **F** (*Waal (NL)*, *Iranian 20x10x5* or a size like *230x110x50 j10/12*) and a bond in **B** (*Stretcher*, *English*, *Flemish*, *Stack*).
3. For openings, connect intervals to **OU** (along the wall) and **OV** (height): use **Construct Domain** or a Panel with *1.2 To 2.4*.
4. Connect **W** to Brick Output, or to an operation.



4 m × 2.4 m Waal wall, stretcher bond, window OU 1.2 To 2.4, OV 0.9 To 2.0. Light bricks are cut.

Tips

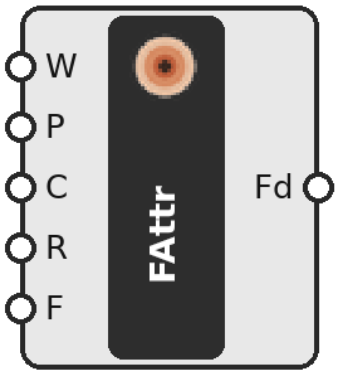
- A single slider in OU/OV becomes "0 To value", so the hole starts at the wall's corner. Always give a start and an end.
- Opening heights starting at 0 are doors.

Field Attractor

جاذب

Hezarbaaf › 2 Design

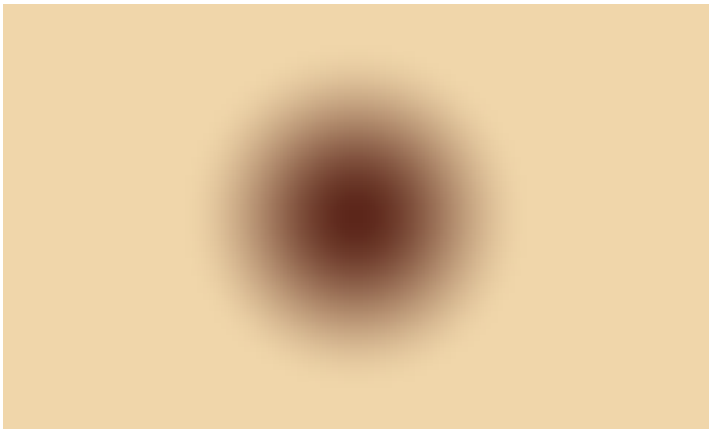
A field is a value from 0 to 1 over the wall. Fields tell operations where to act and how strongly. The attractor field is 1 at points or curves you draw on the wall and fades to 0 at the radius.



Input	What it is	Default
W Wall	Wall base curve (same as Brick Wall)	—
P Points	Attractor points (on/near the wall)	—
C Curves	Attractor curves drawn on the wall	—
R Radius	Influence radius (model units)	1.0
F Falloff	0 linear · 1 smooth · 2 gaussian	1
Output	**What it is**	
Fd Field	Field (0..1 on the façade)	

Try it

1. Connect the same base curve as the wall to **W**: the field needs it to find positions along the wall.
2. Add one or more points to **P** (place them on or near the wall), or curves drawn on the wall to **C**.
3. Set the radius **R** (model units). **F** chooses the fade: 0 linear, 1 smooth, 2 gaussian.
4. Connect **Fd** to Brick Operation, Brick Tones, Glass Bricks or Field Combine.



Attractor at the middle of a 4 × 2.4 m wall, radius 1 m. Dark = 1, light = 0.

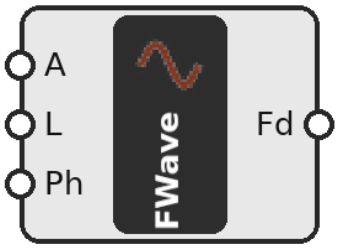
Tips

- Drag the point in Rhino and watch the wall update live.

Field Wave

موج

Straight sine waves across the wall: the classic parametric brick pattern. No wall curve is needed, because waves are measured in façade coordinates.



Input	What it is	Default
A Angle	Direction in degrees (0 = along wall, 90 = upward)	0
L Wavelength	Wavelength (model units)	1.0
Ph Phase	Phase in radians — animate this	0
Output	**What it is**	
Fd Field	Field (0..1 on the façade)	

Try it

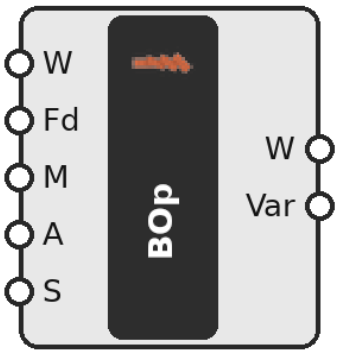
1. Set the wavelength **L** (model units).
2. Set the direction **A** in degrees: 0 = waves change along the wall, 90 = upward, 35 = diagonal.
3. Animate **Ph** (phase) with a slider to slide the waves.



Wave field, L = 1 m, angle 0.

Brick Operation عمليات آجر

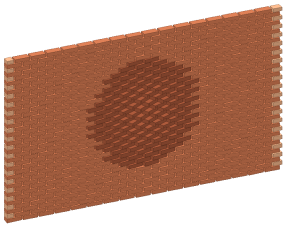
The main design tool. Uses a field to change bricks: **rotate** them, push them **out or in**, **remove** them, or **perforate** (checkerboard holes that always stay supported). Steps keep the result buildable: a mason only needs a few positions.



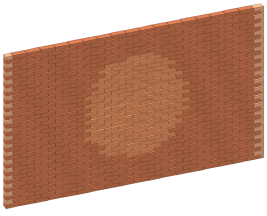
Input	What it is	Default
W Wall	Wall from Brick Wall / Hezarbaaf Wall / another operation	—
Fd Field	Field	—
M Mode	0 rotate · 1 protrude · 2 remove · 3 perforate checkerboard, always supported	0
A Amount	Rotate: max degrees · Protrude: max distance (model units, − = inward) · Remove: threshold 0..1	30
S Steps	Buildable steps (0 = smooth)	4
Output	**What it is**	
W Wall	Resulting wall	
Var Variants	Distinct brick variants (fewer = easier to build)	

Try it

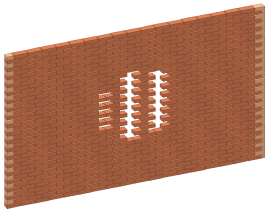
1. **W** ← a wall. **Fd** ← a field.
2. **M**: 0 rotate · 1 protrude · 2 remove · 3 perforate.
3. **A**: rotate = maximum degrees (e.g. 30) · protrude = maximum distance in model units (e.g. 0.03; negative = inward) · remove/perforate = threshold 0–1.
4. **S**: number of steps (4 is a good start; 0 = smooth).
5. Chain several operations: the W output goes into the next operation.



Mode 0, rotate 30° with an attractor field, 4 steps. Turned bricks are darker.



Mode 1, protrude 30 mm



Mode 3, perforate

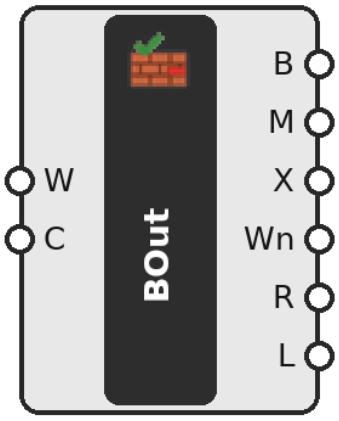
Tips

- **Var** tells you how many different brick positions the wall has: fewer = easier to build.
- Rotated bricks need wider head joints, or they collide (red). Use a format like *210x100x50 j25/12*, or Quick Façade, which widens joints automatically.

Brick Output

خروجی آجر

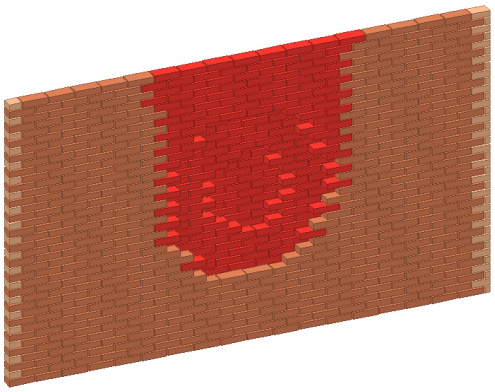
Shows the wall in colour, checks that it can be built, and gives the quantities. Red bricks have a problem: not enough support, too much overhang, or a collision.



Input	What it is	Default
W Wall	Wall	—
C Colour	Preview colours: 0 check (problems red, cut, turned, tapered…) · 1 material (tones, glaze, glass) · 2 plain	0
Output	**What it is**	
B Bricks	Brick boxes	
M Mesh	One coloured mesh (also previewed by this component)	
X Problems	Bricks with buildability problems	
Wn Warnings	Problem descriptions	
R Report	Quantities	
L Legend	What the colours mean	

Try it

1. Connect any wall to **W**.
2. **C** (colour): 0 = check colours (red = problem, light = cut, dark = turned, orange = projecting, purple = tapered, light blue = glass) · 1 = material (tones, glaze, glass) · 2 = plain.
3. Read **Wn** for what each problem is, and **R** for the quantities.
4. Connect **L** to a Panel for the colour legend.



Protrusion pushed too far (16 cm): the steps between courses are too big, so those bricks turn red.

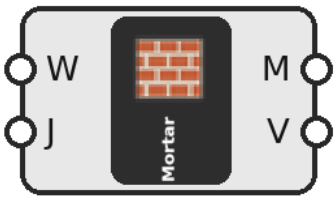
Tips

- Only the last component in a chain draws its wall, so you see one wall, not several overlapping ones.

Mortar

ملات

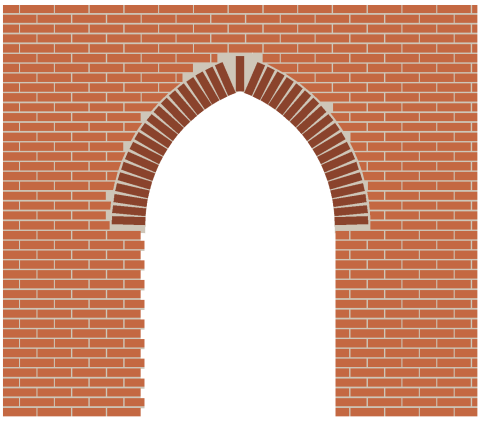
The mortar between the bricks as its own mesh, separate from the bricks, so you can bake it on its own layer and give it its own render material. It fills every joint, including the joint around arches, but never a hole left by a removed or perforated brick, and it never sticks out past wall ends or into openings.



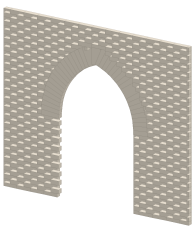
Input	What it is	Default
W Wall	Wall data	—
J Joint	0 flush · 1 recessed (8 mm) · 2 deep (10 mm)	1
Output	**What it is**	
M Mortar	Mortar mesh (bake it, give it a material)	
V Volume	Mortar volume in m³ (joints only, from Quantities)	

Try it

1. Connect any wall to **W**.
2. **J** joint finish: 0 flush · 1 recessed (8 mm back from the face) · 2 deep (10 mm, strong shadow lines). Recessed joints get a weather note: not for walls exposed to driving rain or frost.
3. Bake **M** (or use Bake Blocks, which bakes the mortar on its own "Mortar" layer) and give it a mortar material.
4. **V** gives the mortar volume in m³.



Real mortar geometry behind a wall with a pointed arch (front view): joints, the ring and the crown are all filled.



The mortar mesh on its own, as it bakes

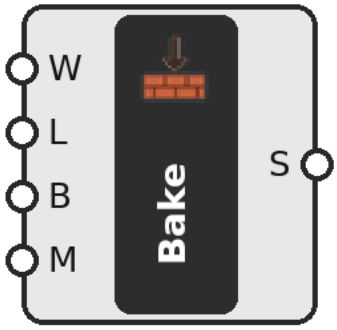
Tips

- Recessed joints only show their shadow in a render: give the mortar a matt, light material and the bricks a rougher one.

Bake Blocks

پخت آجرها در راینو

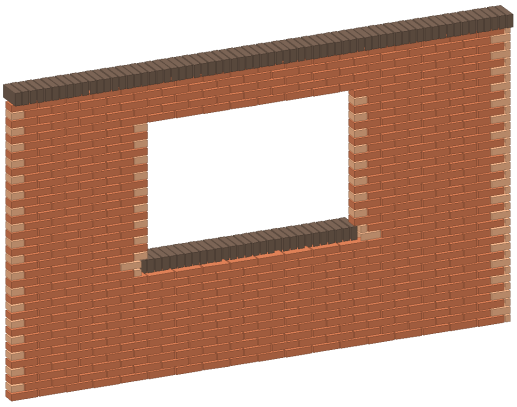
Puts the bricks into Rhino as **block instances**: one block per brick size, one layer per colour (tones, glaze, glass), and the mortar as one mesh on its own **Mortar** layer. Light files, easy to count, and one Undo removes everything.



Input	What it is	Default
W Wall	Wall data	—
L Layer	Layer name prefix	Hezarbaaf
B Bake	Connect a Button	false
M Mortar	Also bake mortar on its own layer: 0 no · 1 flush · 2 recessed · 3 deep	2
Output	**What it is**	
S Status	Result	

Try it

1. Connect a wall to **W**, and set a layer name prefix in **L**.
2. **M** mortar: 0 no · 1 flush · 2 recessed (default) · 3 deep.
3. Connect a **Button** to **B** and click it once.
4. Read **S** for how many blocks were made.



Everything you see in the preview is what gets baked.

Tips

- Bake again after changes; old blocks stay until you delete them.

Part 2 · Walls

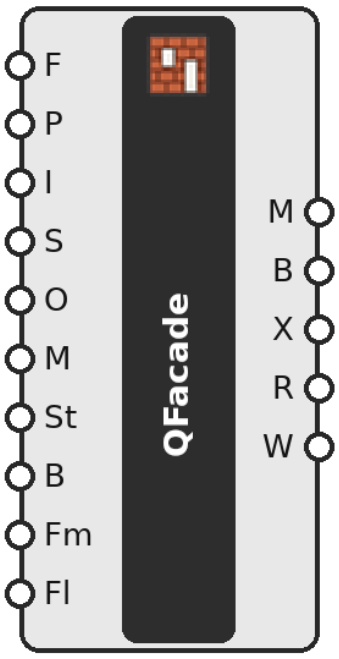
8 components: Quick Façade, Corner Wall, Loft Wall, Section Wall, Surface Wall, Tower, Arched Openings, Details

Quick Façade

نمای سریع

Advanced › 1 Walls

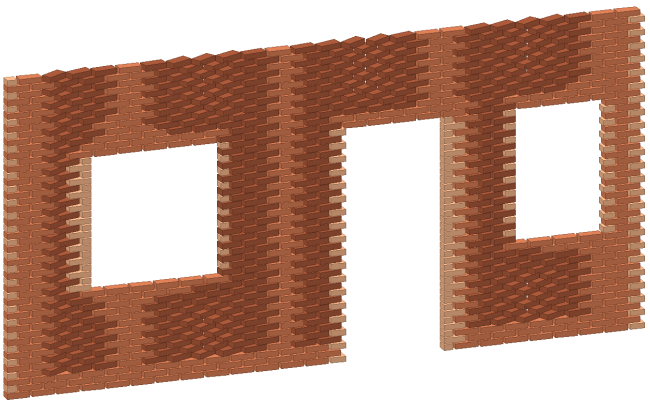
The beginner's shortcut: select a flat façade surface with holes for windows and doors, move a few sliders, and get a complete parametric brick façade. Edges and the area around openings stay calm automatically, and joints widen where bricks turn.



Input	What it is	Default
F Façade	Flat, vertical façade surface; holes = windows/doors	—
P Pattern	0 none · 1 wave · 2 ripple · 3 noise · 4 diagonal wave	1
I Intensity	0..1 rotation strength	0.6
S Scale	Pattern size (model units; default 1.5 m)	—
O Openness	0..1 perforation	0
M Margin	Calm zone around openings (model units; default 0.4 m)	—
St Steps	Buildable steps (0 = smooth)	4
B Bond	Bond	Stretcher
Fm Format	Brick format	Waal (NL)
Fl Flip	Flip outside direction	false
Output	**What it is**	
M Mesh	Bricks as one mesh (fast)	
B Bricks	Brick boxes	
X Problems	Bricks with buildability problems	
R Report	Quantities	
W Wall	Wall data → Brick Operation / Brick Output	

Try it

1. Draw a vertical planar surface for the façade and trim the windows and doors out of it. Connect it to **F**.
2. **P** pattern: 0 none · 1 wave · 2 ripple · 3 noise · 4 diagonal wave. **I** = intensity 0–1.
3. **S** = pattern size, **O** = openness (perforation), **M** = calm margin around openings.
4. If the bricks face inward, set **Fl** (flip) to True.



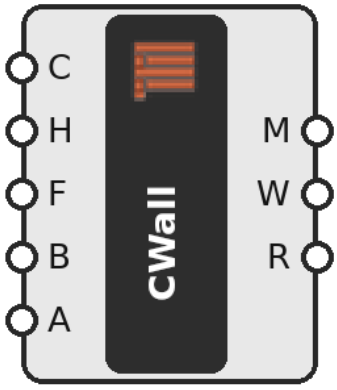
6 × 3 m façade with two windows and a door, wave pattern, intensity 0.7.

Corner Wall

دیوار با کنج

Advanced › 1 Walls

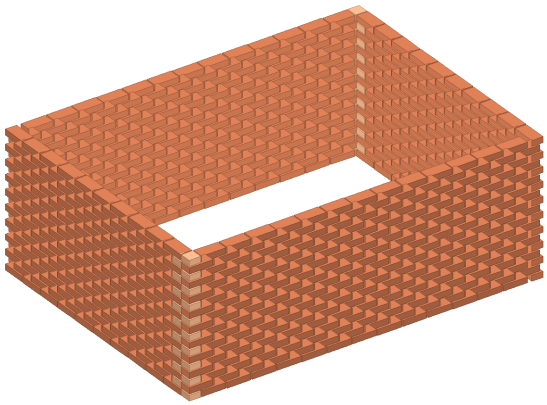
A wall along a polyline with proper corners: at every sharp turn, the two walls take turns running through the corner, course by course, like real masonry. A closed polyline gives a room.



Input	What it is	Default
C Curve	Plan polyline (open or closed)	—
H Height	Wall height	2.0
F Format	Brick format, or a size like 230x110x50 j10/12	Waal (NL)
B Bond	Bond name, or text like S:0 | S:-0.5	Stretcher
A Angle	Minimum turn (degrees) that counts as a corner	30
Output	**What it is**	
M Mesh	Bricks as one mesh	
W Wall	Wall data	
R Report	Quantities + problems	

Try it

1. Draw a polyline (or a closed rectangle for a room) and connect it to **C**.
2. Set **H**, **F** and **B** like Brick Wall.
3. **A**: turns sharper than this angle count as corners (default 30°).



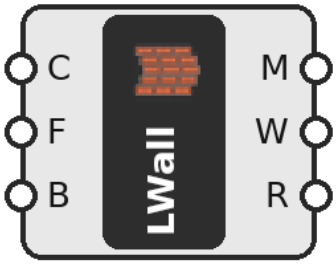
3 × 2.2 m room, 1.2 m high: the corners interlock course by course.

Loft Wall

دیوار از منحنی‌ها

Advanced › 1 Walls

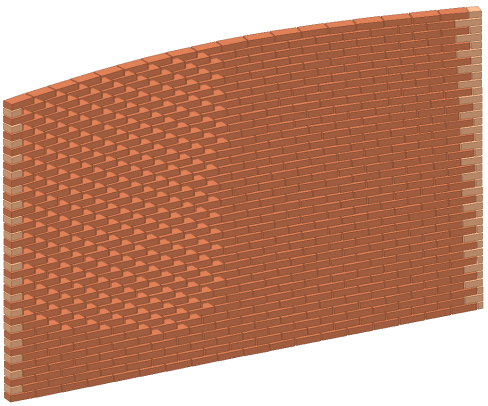
A free-form wall between 2 or 3 guide curves (bottom, middle, top). Bricks stay level; each course follows the interpolated curve, so the wall leans and bends.



Input	What it is	Default
C Curves	Guide curves (bottom, middle…, top), same direction	—
F Format	Brick format	Waal (NL)
B Bond	Bond	Stretcher
Output	**What it is**	
M Mesh	Bricks as one mesh	
W Wall	Wall data	
R Report	Quantities + problems	

Try it

1. Draw 2–3 curves at different heights, all running in the same direction. Connect them in order (bottom first) to **C**.
2. Set **F** and **B**. Red bricks mean a course sticks out too far over the one below.



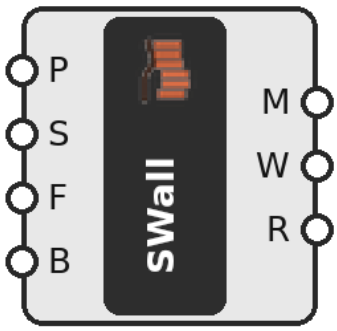
Straight bottom, wavy top: the wall twists between them.

Section Wall

دیوار با مقطع

Advanced › 1 Walls

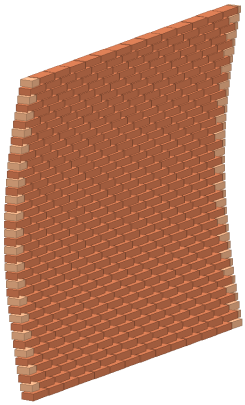
A wall with a curved cross-section: a plan curve plus a profile (how far the wall moves in or out at each height), like a bulging or leaning wall.



Input	What it is	Default
P Plan	Plan curve	—
S Section	Section profile: X offset vs Z height, from its start point	—
F Format	Brick format	Waal (NL)
B Bond	Bond	Stretcher
Output	**What it is**	
M Mesh	Bricks as one mesh	
W Wall	Wall data	
R Report	Quantities + problems	

Try it

1. Connect the plan curve to **P**.
2. Draw the profile in the Front view: X = offset, Z = height, starting at its start point. Connect it to **S**.



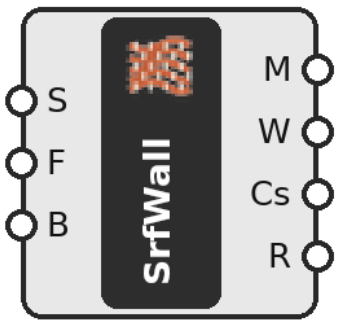
3 m wall bulging 15 cm at mid-height.

Surface Wall

دیوار روی سطح

Advanced › 1 Walls

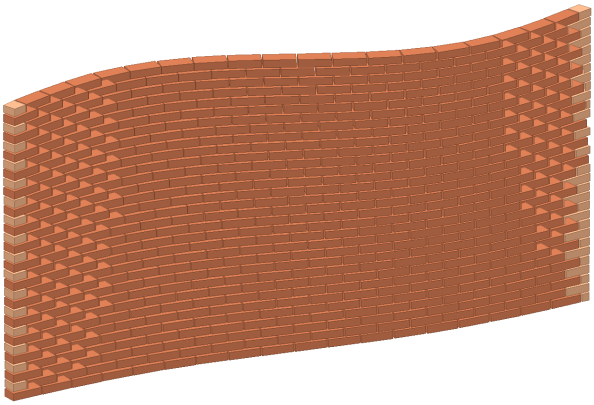
Bricks on any surface or polysurface. The surface is sliced into one contour per course; holes become openings (the bond continues across them), and leaning parts become stepped courses (corbelling), checked like any wall.



Input	What it is	Default
S Surface	Wall surface (the brick centre-line plane)	—
F Format	Brick format, or 230x110x50	Waal (NL)
B Bond	Bond name or text	Stretcher
Output	**What it is**	
M Mesh	Bricks	
W Wall	Wall data	
Cs Courses	Course lines (for setting out)	
R Report	Quantities + problems	

Try it

1. Connect a surface (the brick centre-line) to **S**.
2. Set **F** and **B**. **Cs** gives the course lines for setting out on site.



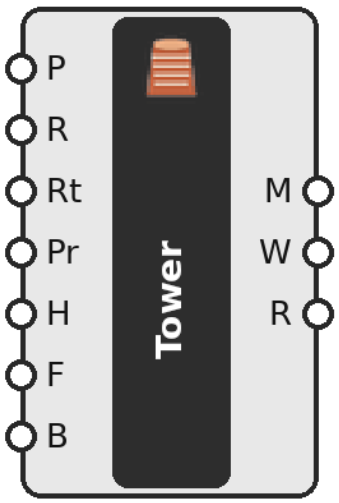
Wavy surface whose waves grow towards the top.

Tower

برج و ستون

Advanced › 1 Walls

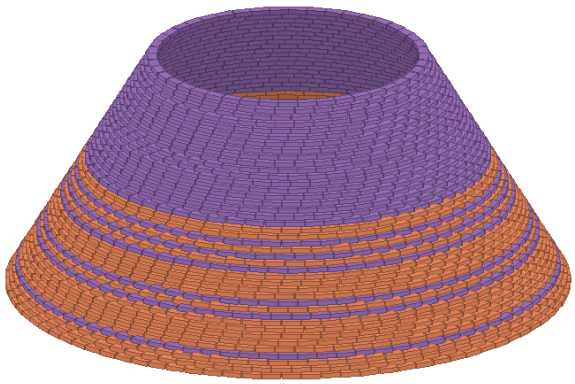
Round towers, minarets, columns and dome profiles. Every ring closes with whole bricks (joints adjust slightly). Where the curve is too tight for normal bricks, they become **tapered**: shown as wedges, and listed in the Raj sheet with how much to cut.



Input	What it is	Default
P Centre	Base centre	origin
R Radius	Base radius to the wall centre	1.5
Rt Top radius	Radius at the top (smaller = minaret taper)	1.5
Pr Profile	Optional profile (Front view: X = radius, Z = height) — overrides R/Rt	—
H Height	Height	4
F Format	Brick format	Iranian 20x10x5
B Bond	Bond (Stretcher, English, Flemish, Header text 'H:0 | H:-0.5h')	Stretcher
Output	**What it is**	
M Mesh	Bricks (tapered ones as wedges)	
W Wall	Wall data → operations, Raj sheet, Bake	
R Report	Tapered bricks, joints, minimum radius	

Try it

1. Set the centre **P**, base radius **R** and top radius **Rt** (smaller = minaret taper), and height **H**.
2. For a dome or badgir shape, draw a profile in the Front view (X = radius, Z = height) and connect it to **Pr**.
3. The **R** output (Report) tells you how many tapered bricks you need and the smallest radius for standard bricks.



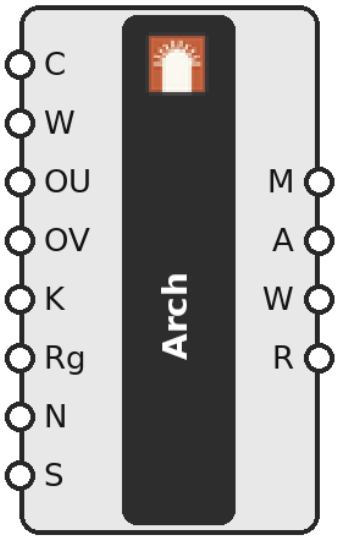
Tower from Ø6 m at the base to Ø2.9 m at the top, Iranian bricks. The upper, tighter rings need tapered bricks (purple).

Arched Openings

طاق و درگاه

Advanced › 1 Walls

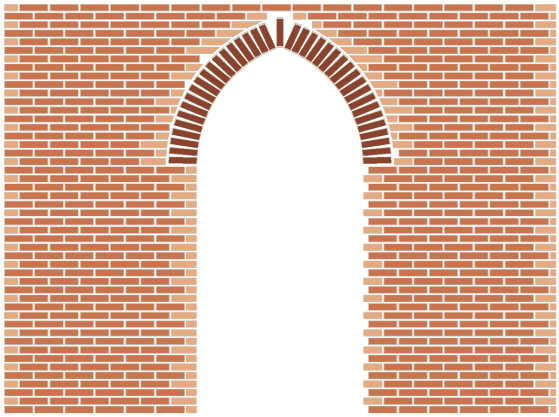
Openings with real brick arches: semicircular, segmental, two-centred pointed (جناغی), Persian panj-o-haft (پنجوهفت) or a flat soldier lintel. Joints are checked against 3–19 mm and the ring depth against span/12 (BIA TN 31). The wall bricks beside the arch are cut on the slant to follow its back (one mortar joint between them, angles listed in the Raj sheet), and the bricks above count as carried by it. The report gives joint widths and warns when tapered voussoirs are needed or when the arch is taller than the wall.



Input	What it is	Default
C Curve	The wall's base curve (same as Brick Wall)	—
W Wall	Plain wall (no openings yet)	—
OU Along	Opening width along the wall, one interval per opening (model units)	—
OV Height	Sill → springing line (top of the jambs)	—
K Kind	0 semicircular · 1 segmental · 2 pointed · 3 flat soldier · 4 panj-o-haft	2
Rg Ring	0 rowlock: brick length to the centre · 1 header: brick width to the centre	0
N Rings	Number of rings 1–3	1
S Shape	Segmental: rise (model units, 0 = span/6) · Pointed: sharpness 0.1–0.5 · Panj-o-haft: base angle 30–70° (default 50)	0.25
Output	**What it is**	
M Mesh	Wall bricks	
A Arches	Arch bricks	
W Wall	Wall data with arches	
R Report	Per arch: bricks, joints, tapered?	

Try it

1. Make a plain wall (Brick Wall **without** OU/OV) and connect it to **W**, with the same curve in **C**.
2. **OU** = width along the wall, **OV** = sill To springing line (top of the jambs).
3. **K**: 0 semicircular · 1 segmental · 2 pointed · 3 soldier lintel · 4 panj-o-haft (Persian four-centred). **Rg**: 0 rowlock, 1 header. **N** = rings.
4. **S**: pointed = sharpness 0.1–0.5; segmental = rise; panj-o-haft = base arc angle 30–70° (default 50).



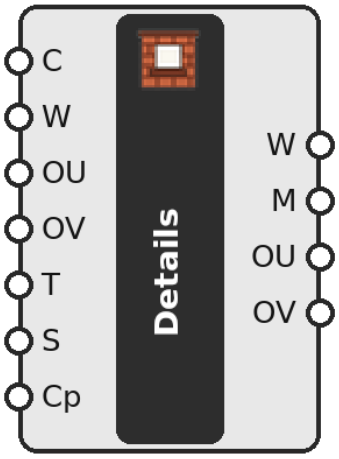
1.2 m Persian pointed arch, door height 1.8 m to the springing line. A key brick fills the apex.

Details

جزئيات

Advanced › 1 Walls

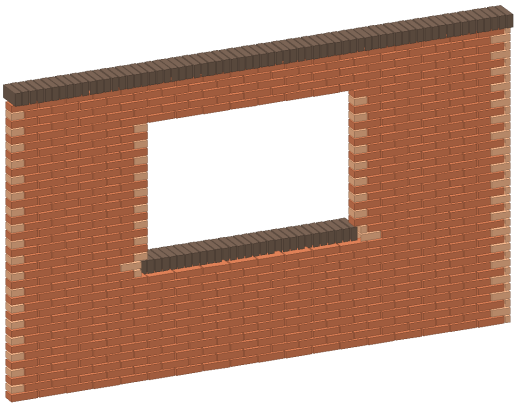
Makes openings look finished: jambs end in whole or half bricks (no slivers), a brick-on-edge sill under each window, and a coping along the top.



Input	What it is	Default
C Curve	The wall's base curve	—
W Wall	Plain wall (no openings yet)	—
OU Along	Opening widths along the wall	—
OV Height	Opening heights (start at 0 = door)	—
T Tidy	Snap jambs to whole/half bricks	true
S Sills	Brick sills under windows	true
Cp Coping	Brick-on-edge coping on top	true
Output	**What it is**	
W Wall	Wall with details	
M Mesh	Bricks	
OU Along	Openings as built (after tidying)	
OV Height	Openings as built	

Try it

1. Plain wall in **W**, same curve in **C**, openings in **OU**/**OV**.
2. Switch **T** (tidy jambs), **S** (sills) and **Cp** (coping) on or off.
3. The **OU**/**OV** outputs give the openings as built (slightly moved by tidying), to use for the window frames.



Window with sill, tidy jambs and coping.

Part 3 · More fields

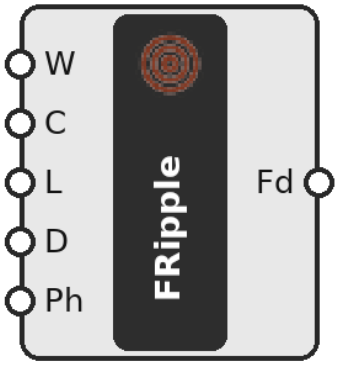
5 components: Field Ripple, Field Noise, Field Bitmap, Field Combine, Field Preview

Field Ripple

موج دایره‌ای

Advanced › 2 Fields

Circular waves from a centre point, like a stone dropped in water.



Input	What it is	Default
W Wall	Wall base curve	—
C Center	Ripple centre	—
L Wavelength	Distance between rings	0.6
D Decay	Fade distance, 0 = none	0
Ph Phase	Phase (animate)	0
Output	**What it is**	
Fd Field	Field (0..1 on the façade)	

Try it

1. Wall curve in **W**, centre point in **C**, ring spacing in **L**.
2. **D** fades the rings with distance (0 = no fade). Animate **Ph**.



Ripple from the middle, rings 0.6 m apart.

Field Noise

نویز

Advanced › 2 Fields

Smooth random variation, which makes a wall look hand-made and lively. Same seed = same pattern.



Input	What it is	Default
S Scale	Feature size (model units)	0.5
Sd Seed	Random seed	0
Output	**What it is**	
Fd Field	Field (0..1 on the façade)	

Try it

1. **S** = feature size (model units). **Sd** = seed; change it for a different pattern.



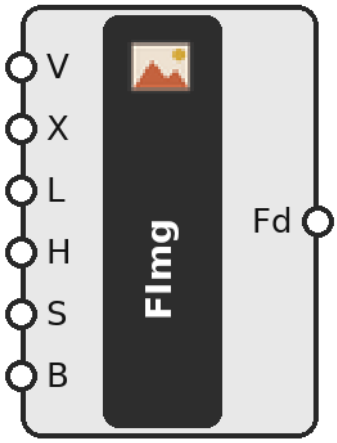
Noise, scale 0.5 m.

Field Bitmap

تصوير

Advanced › 2 Fields

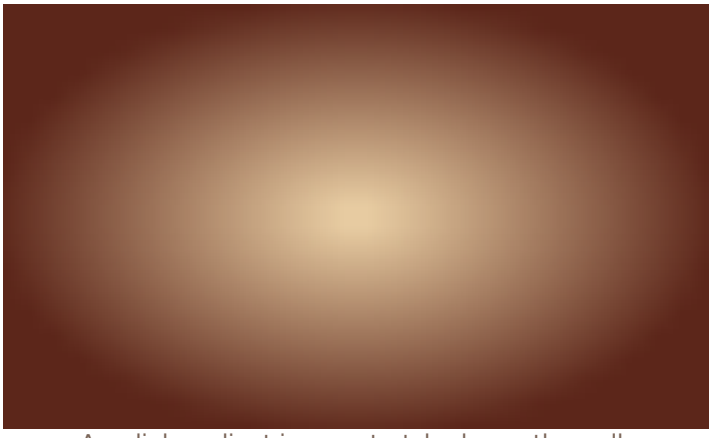
Turns an image into a field: portraits, logos, calligraphy or gradients onto the wall.



Input	What it is	Default
V Values	Brightness 0..1, row by row from the top	—
X Columns	Pixels per row	—
L Length	Image width on the wall (model units)	—
H Height	Image height on the wall	—
S Start	Offset along the wall	0
B Bottom	Offset from the bottom	0
Output	**What it is**	
Fd Field	Field (0..1 on the façade)	

Try it

1. Use Grasshopper's **Image Sampler** (Params › Input) on a picture and connect its values to **V**, row by row from the top.
2. Set **X** = pixels per row (the sampler's X count).
3. Set where the image sits on the wall: width **L**, height **H**, start **S**, bottom **B**.



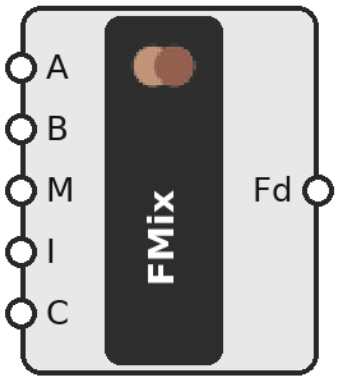
A radial gradient image stretched over the wall.

Field Combine

ترکیب میدان

Advanced › 2 Fields

Mixes two fields, or adjusts one. For example: wave × attractor (a wave only near a point), or invert a field.



Input	What it is	Default
A A	Field A	—
B B	Field B (optional)	—
M Mode	0 add · 1 multiply · 2 max · 3 min · 4 subtract · 5 B modulates wave A (bitmap→wave)	1
I Invert	Invert result	false
C Contrast	1 = unchanged	1
Output	**What it is**	
Fd Field	Field (0..1 on the façade)	

Try it

1. Fields in **A** and **B** (B optional).
2. **M**: 0 add · 1 multiply · 2 max · 3 min · 4 subtract · 5 B modulates the waves of A.
3. **I** inverts the result, **C** changes contrast.



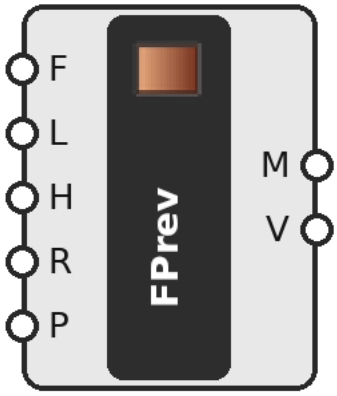
Wave × attractor: the wave fades out away from the point.

Field Preview

پیش‌نمایش میدان

Advanced › 2 Fields

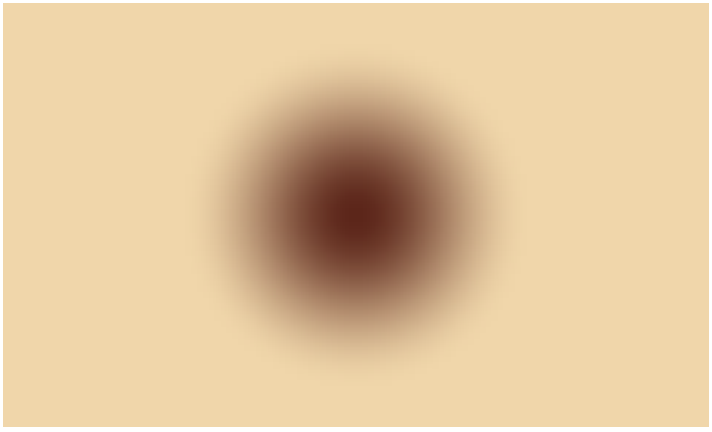
Shows any field as a coloured heatmap on a flat rectangle before you lay a single brick. Good for tuning sizes and radii quickly.



Input	What it is	Default
F Field	Field	—
L Length	Wall length	10
H Height	Wall height	3
R Resolution	Cells along the length	80
P Plane	Façade plane (X along, Y up)	XZ (façade)
Output	**What it is**	
M Mesh	Coloured mesh	
V Values	Sampled values	

Try it

1. Field in **F**, wall size in **L** and **H**, detail in **R**.
2. **P** places the preview (default: XZ plane, like a façade).



Same colours as in this guide: light = 0, dark = 1.

Part 4 · Colour and material

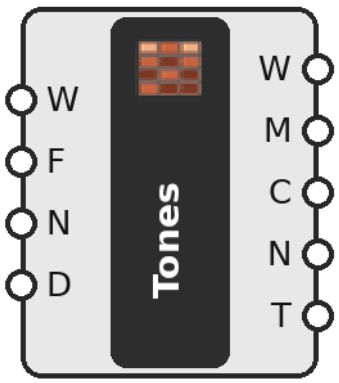
2 components: Brick Tones, Glass Bricks

Brick Tones

رنگ‌آمیزی آجر

Advanced › 3 Operations

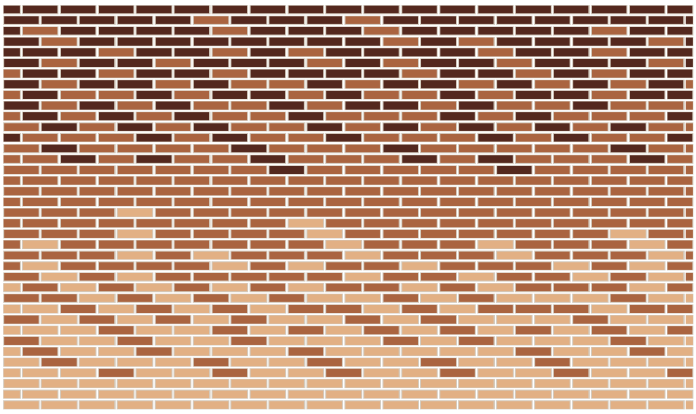
Turns a field into a mix of 2–6 real brick colours you can order. With dithering, the colours mix brick by brick, so three tones read as a smooth gradient from a distance. Arches, sills, copings and glass keep their own look.



Input	What it is	Default
W Wall	Wall data	—
F Field	Field (0 = tone 1, 1 = last tone)	—
N Tones	Number of brick colours (2–6)	3
D Dither	Spread the mix (on) or hard bands (off)	true
Output	**What it is**	
W Wall	Wall data with tones	
M Meshes	One mesh per tone	
C Colours	Preview colour per tone (use Custom Preview)	
N Counts	Bricks per tone	
T Tone	Tone per brick	

Try it

1. Wall in **W**, field in **F** (0 = tone 1, 1 = last tone), number of colours in **N**.
2. **D** on = spread mix, off = hard bands.
3. **Counts** gives the order quantities per colour. In Brick Output, set C = 1 to see the tones.



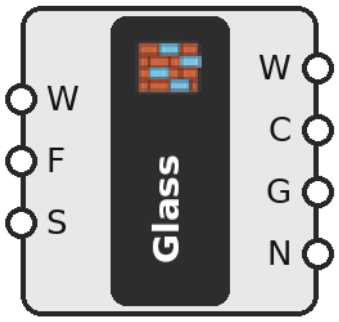
Three tones, light at the base, dark at the top.

Glass Bricks

آجر شیشه‌ای

Advanced › 3 Operations

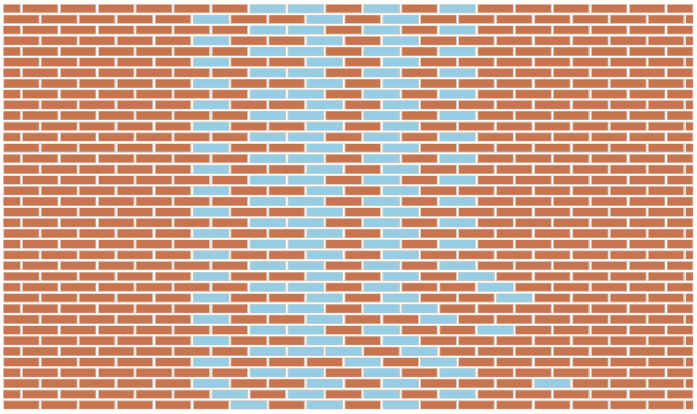
Mixes glass bricks into the wall. The field is the share of glass: 0.3 means about every third brick there. Glass bricks keep the bond, and are counted and priced separately.



Input	What it is	Default
W Wall	Wall data	—
F Field	Share of glass: 0 = clay, 1 = all glass	—
S Spread	On: mix evenly (30 % = every third brick) · Off: hard edge at 50 %	true
Output	**What it is**	
W Wall	Wall with glass bricks	
C Clay	Clay bricks	
G Glass	Glass bricks (use a transparent material)	
N Count	Glass bricks to order	

Try it

1. Wall in **W**, field in **F**. **S** on = mix evenly; off = hard edge at 50 %.
2. Give the **G** mesh a transparent material, or use Brick Output with C = 1.



Up to 60 % glass in the middle of the wall, fading to the sides.

Part 5 · Hezarbaaf motifs

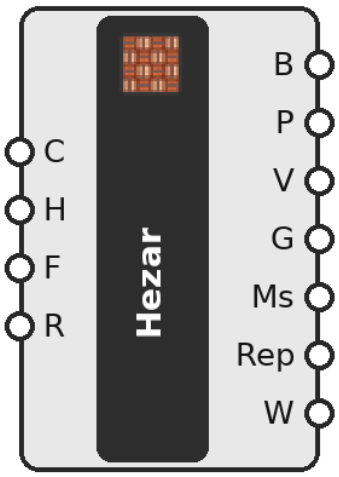
3 components: Hezarbaaf Wall, Banna'i Motif, Image Motif

Hezarbaaf Wall

دیوار هزارباف

Advanced › 4 Hezarbaaf

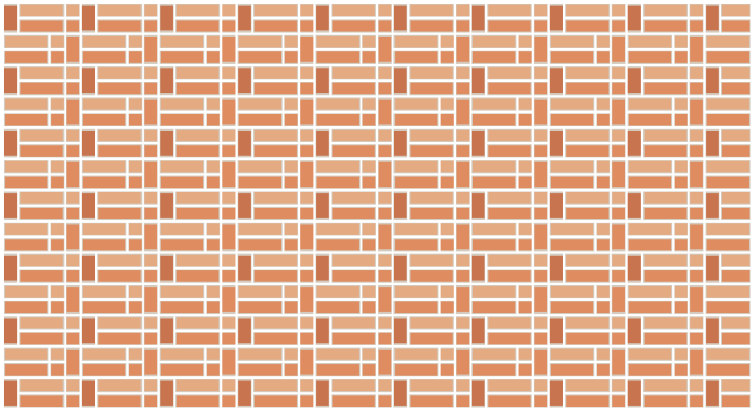
The Persian hezarbaaf technique: woven-looking relief brickwork from a motif you write like pixels. Uppercase letters are flat bricks, lowercase are upright; A flush, B projecting, C recessed, G gold, T turquoise, N navy, "." empty.



Input	What it is	Default
C Curve	Base curve (plan)	—
H Height	Wall height	2.0
F Format	Brick format	Iranian 20x10x5
R Relief	Projection/recess depth	0.02
Output	**What it is**	
B Bricks	Brick pieces	
P Projection	Projection	
V Vertical	Upright brick	
G Glaze	Glaze colour 0-3 (0 = plain)	
Ms Motifs	Preset motifs	
Rep Report	Counts	
W Wall	Wall data → Brick Operation / Brick Output	

Try it

1. Base curve in **C**, height in **H**; Iranian 20×10×5 brick by default.
2. Connect a motif: a preset name (see **Ms**), the output of Banna'i Motif or Image Motif, or your own rows of letters.
3. **R** = relief depth (how far B projects and C recedes).



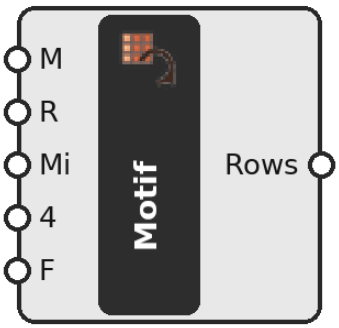
The "Basket (woven)" motif repeated over a 3 × 1.6 m wall.

Banna'i Motif

نقش‌ساز بنایی

Advanced › 4 Hezarbaaf

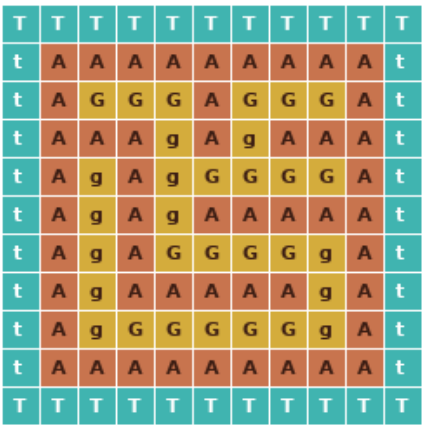
Builds and transforms motifs: rotate in quarter turns, mirror, repeat fourfold around the centre, and add a coloured frame. Square Kufic «علی» is built in.



Input	What it is	Default
M Motif	Preset name or rows	Ali (square Kufic)
R Rotate	Quarter turns 0-3	0
Mi Mirror	Mirror left-right	false
4 Fourfold	Repeat 4× rotated around centre	false
F Frame	Border cell (T, G, N, A…) or empty for none	T
Output	**What it is**	
Rows Rows	Motif rows → connect to Hezarbaaf Wall	

Try it

1. A preset name or your own rows in **M**.
2. **R** quarter turns, **Mi** mirror, **4** fourfold, **F** frame letter (T = turquoise, G = gold…).
3. Connect **Rows** to Hezarbaaf Wall.

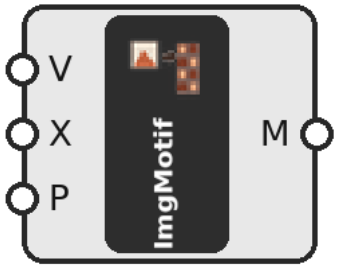


«علی» in square Kufic with a turquoise frame: each cell is one brick.

Image Motif

نقش از تصویر

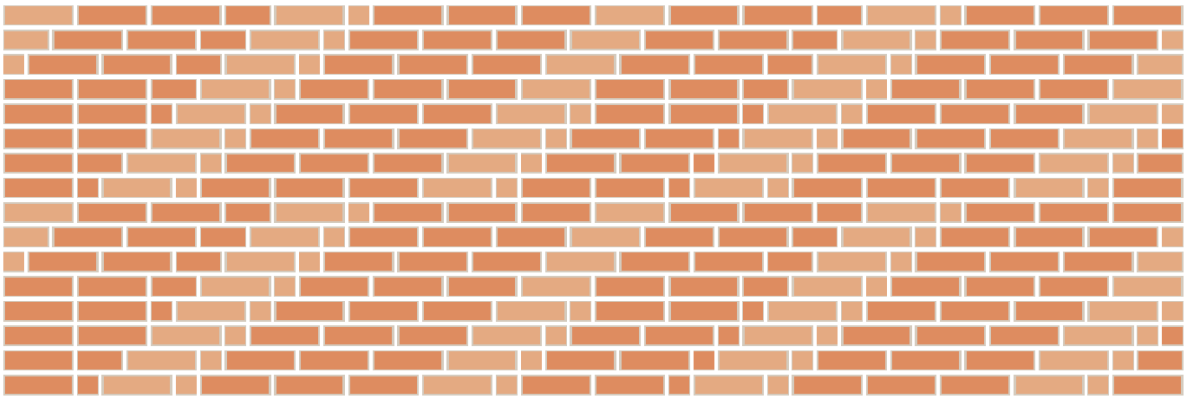
Turns a picture into a Hezarbaaf motif: each pixel becomes a brick cell, and equal neighbours merge into longer bricks.



Input	What it is	Default
V Values	Brightness 0..1, row by row from the top	—
X Columns	Pixels per row	—
P Palette	Letters dark → light. CAB relief · AG gold · .A screen	CAB
Output	**What it is**	
M Motif	Motif rows → Hezarbaaf Wall	

Try it

1. Image Sampler values in **V**, pixels per row in **X**.
2. **P** palette, dark → light: *CAB* relief portrait · *AG* plain/gold · *.A* perforated screen.
3. Connect **M** to Hezarbaaf Wall.



Diagonal gradient image, palette AB: flush and projecting bricks.

Part 6 · From design to site

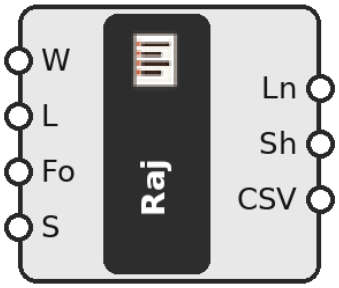
5 components: Raj Sheet, Quantities, Rotation Stencils, Assembly Order, Elevation Drawing

Raj Sheet

رج‌شیت

Advanced › 5 Site

The Raj sheet (رج‌شیت): course-by-course instructions a mason can read aloud on site, plus a CSV list of every brick. Identical neighbouring bricks are grouped, like knitting instructions.



Input	What it is	Default
W Wall	Wall data	—
L Language	fa or en	current language
Fo Folder	Optional folder to save raj-sheet.txt + bricks.csv	—
S Save	Save files (use a button)	false
Output	**What it is**	
Ln Lines	One line per course	
Sh Sheet	Full sheet text	
CSV CSV	Brick schedule	

Try it

1. Wall in **W**, language **fa** or **en** in **L**.
2. To save files, give a folder in **Fo** and click a Button connected to **S**: you get raj-sheet.txt and bricks.csv (opens in Excel).

```
Raj sheet — Hezarbaaf
Bottom to top; in each course from the wall start to its end.
Course 1 (height 0 mm, 14 bricks): 13 stretcher, straight · 1 stretcher, cut 140 mm
Course 2 (height 62 mm, 15 bricks): 1 stretcher, cut 100 mm · 13 stretcher, straight · 1 stretcher, cut 30 mm
Course 3 (height 124 mm, 14 bricks): 13 stretcher, straight · 1 stretcher, cut 140 mm
Course 4 (height 186 mm, 15 bricks): 1 stretcher, cut 100 mm · 13 stretcher, straight · 1 stretcher, cut 30 mm
Course 5 (height 248 mm, 14 bricks): 13 stretcher, straight · 1 stretcher, cut 140 mm
Course 6 (height 310 mm, 15 bricks): 1 stretcher, cut 100 mm · 13 stretcher, straight · 1 stretcher, cut 30 mm
Course 7 (height 372 mm, 9 bricks): 5 stretcher, straight · 1 stretcher, cut 205 mm · 2 stretcher, straight · 1 stretcher, c
Course 8 (height 434 mm, 11 bricks): 1 stretcher, cut 100 mm · 4 stretcher, straight · 1 stretcher, cut 105 mm · 1 stretcher
Course 9 (height 496 mm, 9 bricks): 5 stretcher, straight · 1 stretcher, cut 205 mm · 2 stretcher, straight · 1 stretcher, c
Course 10 (height 558 mm, 11 bricks): 1 stretcher, cut 100 mm · 4 stretcher, straight · 1 stretcher, cut 105 mm · 1 stretche
Course 11 (height 620 mm, 9 bricks): 5 stretcher, straight · 1 stretcher, cut 205 mm · 2
```

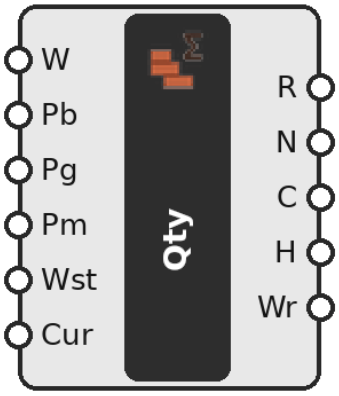
Start of a Raj sheet for a 3 m wall with a window.

Quantities

متره و برآورد

Advanced › 5 Site

Bricks to order (with waste), whole and cut bricks, mortar volume, weight, cost, labour hours, and bed-joint wire (**Wr**) where the bond is stacked or perforated.



Input	What it is	Default
W Wall	Wall data	—
Pb Brick price	Price per brick	0.8
Pg Glazed price	Price per glazed brick	3.0
Pm Mortar price	Price per m³ mortar	150
Wst Waste %	Waste allowance	5
Cur Currency	Currency label	€
Output	**What it is**	
R Report	Detailed report	
N To order	Bricks to order	
C Cost	Total cost	
H Labour h	Person-hours	
Wr Wire	Bed-joint reinforcement for stacked / perforated zones (one curve per joint)	

Try it

1. Wall in **W**; prices in **Pb**, **Pg** (glazed), **Pm** (mortar per m³); waste % in **Wst**; currency label in **Cur**.
2. Outputs: the full report **R**, bricks to order **N**, total cost **C**, person-hours **H**.

```
Bricks: 216 (whole 169, cut 47) · stretchers 216, headers 0 · courses 19
upright 0 · glazed: gold 0, turquoise 0, navy 0
to order (incl. waste): 212 bricks
brick volume 0.20 m³ · mortar 0.05 m³ · weight 0.4 t
cost €178 · labour ≈ 4 person-hours
```

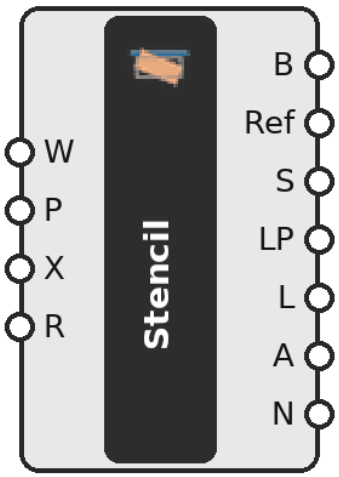
Quantity report for the same wall.

Rotation Stencils

شابلون چرخش

Advanced › 5 Site

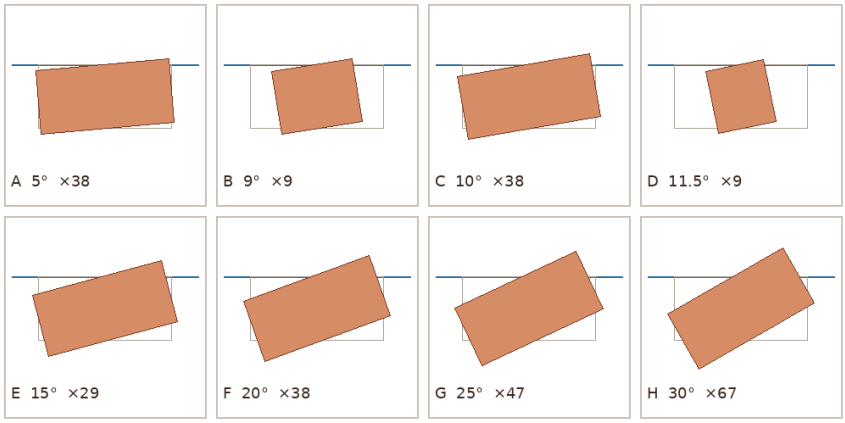
For walls with turned bricks: one printable full-size template per angle. The mason lays the stencil along the string line and turns each brick to match. Letters (A, B, C…) also appear on the elevation drawing.



Input	What it is	Default
W Wall	Wall data	—
P Plane	Where to lay out the sheet	World XY
X Columns	Stencils per row	4
R Round	Angle rounding (degrees)	0.5
Output	**What it is**	
B Brick	Rotated brick outlines	
Ref Reference	Straight brick (print dashed)	
S String line	Wall face line	
LP Label point	Where to put labels (Text Tag)	
L Label	Code · angle · count	
A Angle	Angle per stencil	
N Count	Bricks per stencil	

Try it

1. Wall in **W**; where to lay out the sheet in **P**; stencils per row in **X**.
2. Show **B** (brick), **Ref** (straight brick, print dashed) and **S** (string line); put **L** at **LP** with a Text Tag.
3. Print at 1:1.



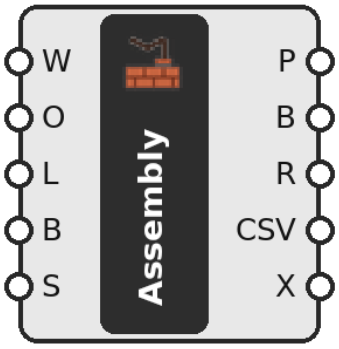
A wave-twisted wall turned in 6 steps needs only 8 stencils (some angles also come as cut bricks).

Assembly Order

ترتیب اجرا

Advanced › 5 Site

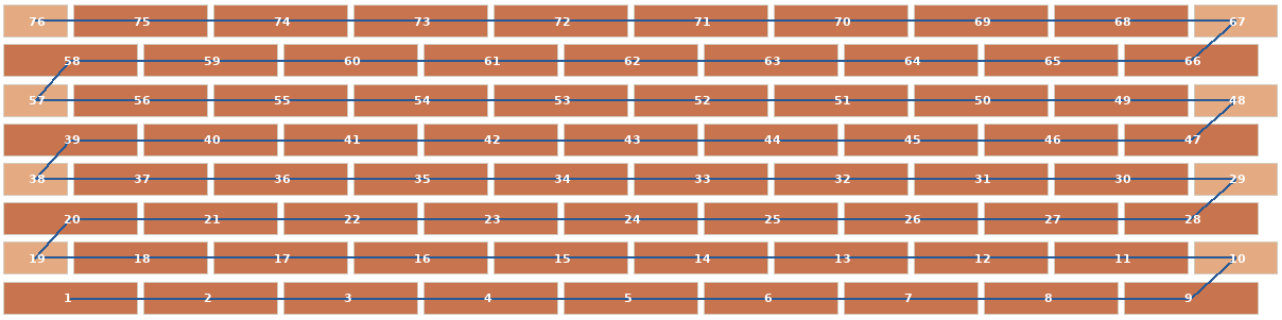
The order to lay the bricks, for a crew or a pick-and-place robot, with a target plane on each brick's top face and a CSV file. It checks that no brick is placed before the bricks it sits on.



Input	What it is	Default
W Wall	Wall data	—
O Order	0 course by course · 1 serpentine (robot) · 2 lifts (scaffold bays)	1
L Lift	Courses per lift (order 2)	6
B Bay	Bay width (order 2, model units)	1.5
S Save to	Optional CSV file path	—
Output	**What it is**	
P Planes	Place planes in order (origin = top face centre)	
B Bricks	Bricks in placing order	
R Rotation	World rotation about Z (degrees)	
CSV CSV	Schedule as CSV	
X Problems	Bricks placed before their support	

Try it

1. Wall in **W**. **O**: 0 course by course · 1 back-and-forth (robot) · 2 lifts (scaffold bays; set **L** courses and **B** bay width).
2. Planes in **P** go to a robot plugin; **CSV** or **S** (file path) for the schedule.



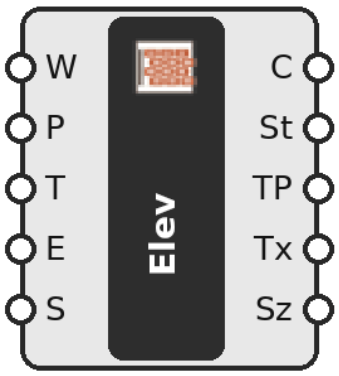
Back-and-forth order: each course runs the other way.

Elevation Drawing

نقشه نما

Advanced › 5 Site

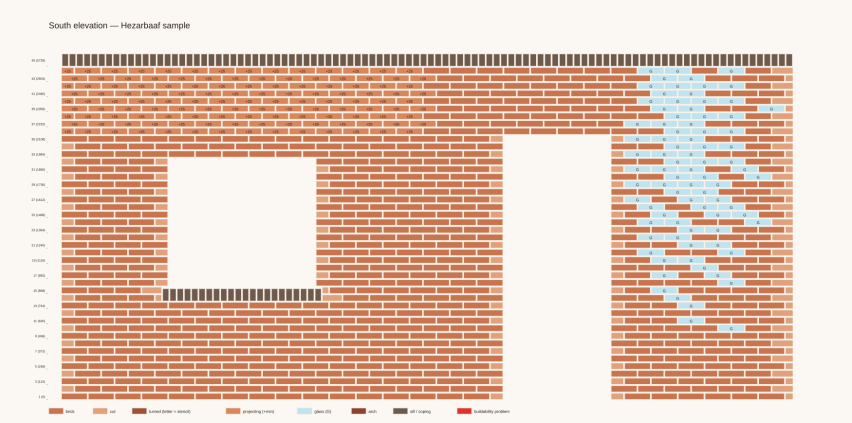
A printable site elevation: every brick, course numbers and heights on the side, stencil letters on turned bricks, ± mm on relief, G on glass, a legend and a title. As Rhino curves and text, or as an SVG file that opens in any browser.



Input	What it is	Default
W Wall	Wall data	—
P Plane	Where to draw it (full size, unrolled)	World XY
T Title	Drawing title	—
E Every	Label every n-th course	1
S Save to	Optional .svg path (opens in any browser, prints to scale)	—
Output	**What it is**	
C Curves	Brick outlines	
St Style	Style per curve (brick, cut, turned, glass, problem…) for colouring	
TP Text planes	Label positions → Text Tag 3D	
Tx Texts	Labels	
Sz Sizes	Text heights	

Try it

1. Wall in **W**; title in **T**; label every n-th course with **E**.
2. To get the file: a path ending in .svg in **S**. Or bake **C** (curves) and use **TP** + **Tx** with Text Tag 3D.



SVG elevation with sill, coping, glass (G) and +25 mm relief.

Part 7 · Ladybug bridge

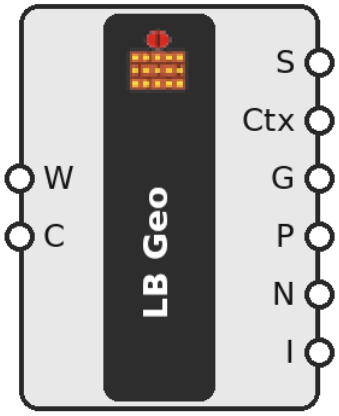
2 components: Ladybug Geometry, Ladybug Results

Ladybug Geometry

برای لیدی‌باگ
هندسه

Advanced › 6 Ladybug

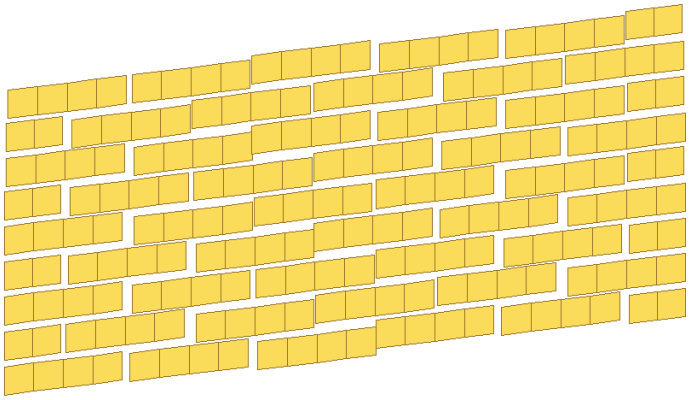
Prepares the wall for a Ladybug sun or radiation study: a **Study** mesh of small sensor faces on the brick fronts, and a **Context** mesh of all bricks so the relief shades itself. Full tutorial: docs/LADYBUG.md.



Input	What it is	Default
W Wall	Wall data	—
C Cell	Sensor face size (model units). 0.05 = about one face per brick height	0.05
Output	**What it is**	
S Study	→ Ladybug _geometry (a Mesh: its faces are used as sensors)	
Ctx Context	→ Ladybug context_ (all bricks, so the relief shades itself)	
G Grid	→ Ladybug Results · Grid	
P Points	Sensor centres	
N Normals	Sensor normals	
I Info	Counts	

Try it

1. Wall in **W**; sensor size in **C** (0.05 m ≈ one row per course).
2. **S** → Ladybug *_geometry*, **Ctx** → *context_*, **G** → Ladybug Results.
3. In LB Incident Radiation set *_offset_dist_* to about 0.002 m (the 10 cm default lifts sensors in front of the relief).



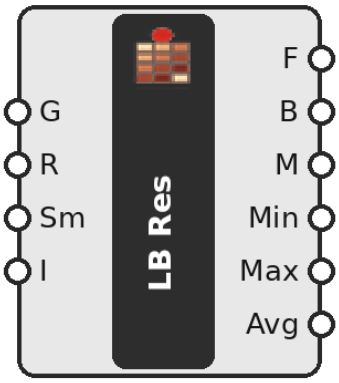
Sensor faces on a relief wall: one Ladybug result per face.

Ladybug Results

نتایج لیدی‌باگ

Advanced › 6 Ladybug

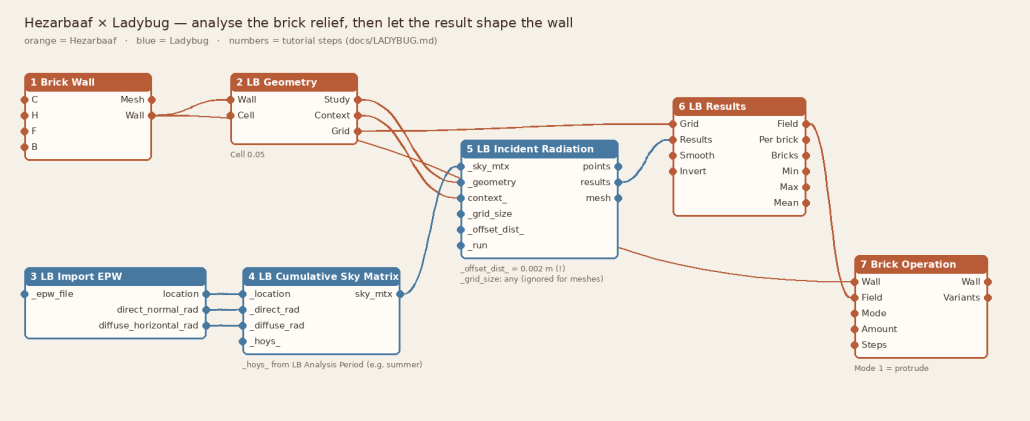
Brings Ladybug's results back onto the bricks: a value per brick, coloured bricks, min/max/mean, and a 0–1 field that can drive any operation. For example, push bricks out where summer sun is strongest.



Input	What it is	Default
G Grid	From Ladybug Geometry · Grid	—
R Results	Ladybug results (one per Study face, flat list)	—
Sm Smooth	Field smoothing radius (model units)	0.15
I Invert	Invert: 1 where the result is lowest	false
Output	**What it is**	
F Field	0..1 field (min → 0, max → 1) → any Hezarbaaf operation	
B Per brick	Mean result per brick (Ladybug units)	
M Bricks	Bricks coloured by result (sand = low, deep clay = high)	
Min Min	Lowest result	
Max Max	Highest result	
Avg Mean	Area-weighted mean over the wall	

Try it

1. **G** ← Ladybug Geometry · G. **R** ← Ladybug results (a flat list).
2. **Sm** smooths the field; **I** inverts it (1 where the result is lowest).
3. **F** → Brick Operation, Brick Tones or Glass Bricks.



The full chain: wall → Ladybug → result field → new wall.

Part 8 · Library

3 components: Brick Formats, Bonds, Language

Brick Formats

اندازه‌های آجر

Advanced › 7 Library

Lists the built-in brick sizes, in English and Persian: Waal, Dik, Vecht and Hilversum (NL), UK standard, US modular and Iranian 20×10×5. Any Format input also accepts a typed size such as *230x110x50 j10/12* (length × width × height, head/bed joint in mm; Persian digits work too).



Input	What it is	Default
Output	**What it is**	
F Formats	Names (English)	
Fa نام فارسی	Persian names	

Try it

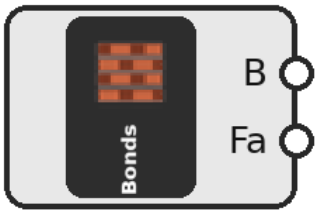
1. Connect **F** or **Fa** to a Panel, or to a Value List to pick from.

Bonds

چینش‌ها

Advanced › 7 Library

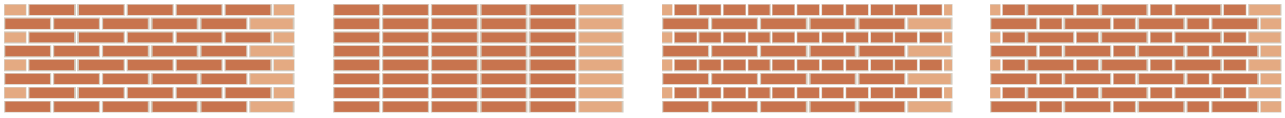
Lists the built-in bonds: Stretcher, Stack, English and Flemish. Any Bond input also accepts a typed bond, one course pattern per "|": S = stretcher, H = header, and the offset after ":" in bricks (e.g. *S:0 | S:-0.25 | S:-0.5 | S:-0.75* for a raking quarter bond).



Input	What it is	Default
Output	**What it is**	
B Bonds	Names (English)	
Fa نام فارسی	Persian names	

Try it

1. Connect **B** or **Fa** to a Panel or Value List.



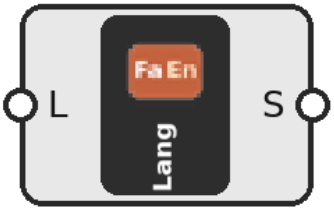
Stretcher Stack English Flemish

Language

زبان

Advanced › 7 Library

Switches Hezarbaaf between English and Persian: component names, tabs, descriptions, reports and Raj sheets.



Input	What it is	Default
L Language	en or fa	en
Output	**What it is**	
S Status	Status	

Try it

1. Type **fa** or **en** in **L**.
2. Restart Rhino: Grasshopper reads the language when it starts.

Troubleshooting

A component is red: "W needs a Hezarbaaf wall…, but a Curve is connected"	You wired a curve into a Wall input. Put Brick Wall (or Presets) in between: Curve → Brick Wall → W.
The opening starts at the corner of the wall	A single slider in OU/OV becomes "0 To value". Use Construct Domain (start, end) or a Panel like *1.2 To 2.4*.
Red bricks	Hover over Brick Output · Wn to read why. "collides": widen the head joint or turn less. "not enough support": the course below is missing or too far away. "overhang": a course sticks out more than 1/3 of the brick depth; use fewer steps or less amount.
Arched Openings warns "the arch rises above the wall"	The crown is taller than the wall. Make the wall taller, or the opening narrower or lower.
Pink or red Grasshopper preview over the bricks	Older components on the canvas still show Grasshopper's default preview. Right-click their mesh outputs › Preview off. New components start hidden.
Bricks face the wrong way	The front of a wall is on the left of the curve direction. Flip the curve (Rhino: Dir / Flip), or use Flip in Quick Façade.
Persian names do not show	Language component: fa, then restart Rhino.
Everything is tiny or huge	Check Rhino's units (File › Properties › Units). Inputs use model units; brick sizes are always in mm.